

TEL26: all-atom vs CG-ML dynamics

Hybrid-2 (3+1) G-quadruplex (2JPZ, (TTAGGG)₄TT), 10 copies × 26 nt. Structure (P(r), g(r), R_g and collective variables), dynamics (translational and rotational MSD, also per GPU hour), a description of the model and a priors-only control. Project PaiNN-LT · MLCG_Framework_v2 · 8 October 2026

Summary

- **The PaiNN residual brings the structure to the AA ceiling.** With the same protocol (2 × 20 ns, same initial state) and the lp2 priors alone, the intramolecular all-sites overlap is 0.980; with PaiNN 0.995, against a ceiling of 0.995. In the B3-B3 channel (Hoogsteen bonds) it goes from 0.841 to 0.976; the core rmsd drops from 0.088 to 0.044 nm (AA 0.044).
- **Core and local structure reproduced within the noise of the reference.** The intramolecular P(r) per channel overlap with the AA ones by 0.958–0.995, against a ceiling (half vs half of the AA trajectory) of 0.961–0.995; core rmsd: overlap 0.93 (ceiling 0.94).
- **Correct mean R_g:** 0.800 nm vs 0.798 nm (AA); overlap 0.86 (ceiling 0.89). The AA copies stay on different R_g values for tens of ns; in the CG all copies fluctuate around the same value.
- **Limitation: the loops.** Loop rmsd: overlap 0.58 (ceiling 0.88), mean 0.272 vs 0.217 nm. The CG has a single basin, whereas the AA has metastable states lasting ns.
- **Efficiency per GPU hour** (A100; 2 replicas per GPU, 54.2 ns/day aggregate, vs 137 ns/day for the AA): translation 137× (MSD); rotation 44× (decorrelation); core rmsd 28×, the minimum among comparable observables.

1. Systems, models and data

	All-atom (reference)	CG-ML
Representation	169 000 atoms, explicit solvent	10 × 86 = 860 beads (G: backbone S + bases B1–B5; T, A: one site), implicit solvent
Hamiltonian	atomistic force field	physical priors lp2 (bonds, angles, dihedrals, 54 Morse contacts per copy, WCA) + PaiNN residual (<i>re1_it30</i> , trained by relative entropy); section 2
Dynamics	dt 2 fs	Langevin, dt 4 fs, γ = 2; ESPResSo + PaiNN plugin
Trajectories	1 × 135.8 ns, 10 copies	2 replicas × 20 ns (4 segments × 5 ns), 10 copies per replica
Analysed frames	6788 frames every 20 ps (P(r): one every 2)	structure: every 20 ps (2008 frames); MSD and times: every 1 ps
Control	–	lp2 priors only (PaiNN off), same 2 × 20 ns chain from the same initial state
Throughput	137 ns/day (1 A100 + 32 cores)	54.2 ns/day aggregate (2 replicas on 1 A100 + 8 cores)

The collective variables (R_g, core and loop rmsd, Q) are computed for each copy after rebuilding the molecule across the periodic boundary, relative to the mean structure of the first half of the AA trajectory. Overlaps between distributions are $\int \min(p_{AA}, p_{CG}) dx$ (1 = identical). The **ceiling** is the overlap between the second and the first half of the AA trajectory: it is the best agreement one can ask for with this reference statistics.

2. The CG-ML model: mapping, priors and PaiNN residual

The potential energy of the model is the sum of two parts: $U(R) = U_{\text{prior}}(R) + U_{\text{ML}}(R)$. U_{prior} contains analytic terms with a physical meaning (connectivity, local geometry, Hoogsteen bonds, stacking, excluded volume), fitted to the mapped all-atom structure at 300 K. U_{ML} is a residual learned with a PaiNN network; it corrects what the priors do not describe. Solvent and ions are implicit: the production model has no electrostatic term.

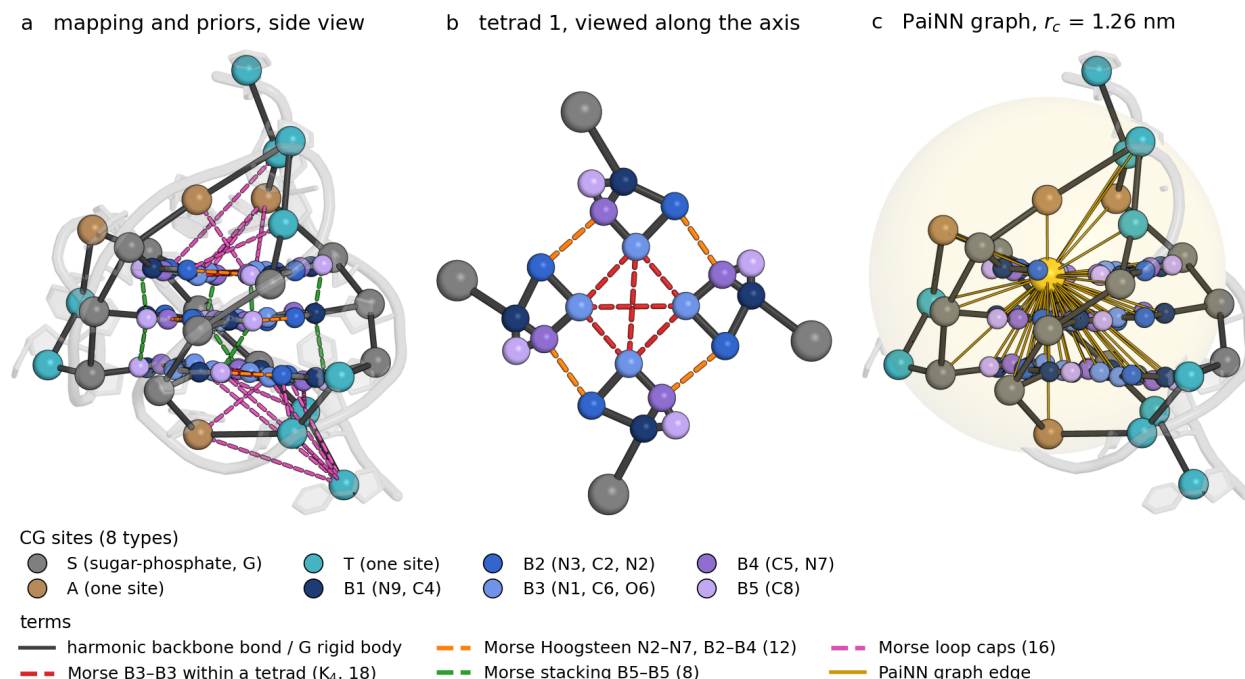


Figure 1. The CG model on the native 2JPZ structure (starting structure of the AA simulations, mapped onto the CG sites; the atomistic DNA in transparent grey). **a**: sites coloured by type; harmonic backbone bonds and the rigid skeleton of the guanines in dark grey; Morse contacts dashed, by class. **b**: tetrad 1 (G4, G12, G16, G22) viewed along the quadruplex axis, with the six B3-B3 Morse terms (sides and diagonals) and the four Hoogsteen B2-B4 Morse terms. **c**: the graph on which PaiNN acts for the B3 site of G16; the sphere has radius $r_c = 1.26$ nm and contains 76 of the other 85 sites of the copy. Angles, dihedrals and WCA are not drawn. Rendered with PyMOL.

2.1 Mapping

Each 26-nucleotide copy is represented by 86 sites of 8 types. Each site is the centre of mass of a group of atoms:

- **guanine**: six sites joined in a **rigid body**. S collects sugar and phosphate (P, O5', C5', C4', O4', C1', C3', C2', O3'). B1-B5 split the base: B1 = N9, C4; B2 = N3, C2, N2; B3 = N1, C6, O6; B4 = C5, N7; B5 = C8. B3 carries the O6 facing the channel, B2 and B4 the donor and acceptor of the second Hoogsteen bond (N2-H...N7), B5 the stacking face;
- **thymine and adenine**: a single site, the centre of mass of the whole nucleotide;
- **dynamics**: Langevin in ESPReso. The 12 guanines are rigid bodies with mass and inertia tensor from the mapping (6 degrees of freedom); A and T are point masses. In total there are 114 degrees of freedom per copy.

Note: the definition of S also lists the non-bridging oxygens under the names O1P/O2P, whereas the force field calls them OP1/OP2. They therefore do not enter the centre of mass, in the same way in the reference and in all datasets.

2.2 Physical priors (set *lp2*)

The parameters come from the mapped AA structure at 300 K (10 copies, 136 ns). Bonds and angles are obtained by Boltzmann inversion. For the Morse terms r_0 is the median distance, and $a = \sqrt{(kT / 2D\sigma^2)}$ uses the observed width σ , later refined iteratively on the distributions of the reference. The contacts are pair-specific: each one binds two given sites of the native fold. At run time they are non-bonded interactions on marker sites, with a tail smoothly switched to zero at r_{cut} , so they can break without "bond broken" errors.

term	form	sites	per copy	parameters
backbone bonds	$\frac{1}{2} k (r - r_0)^2$	site S, A or T of consecutive residues	25	k 136-3755 kJ mol ⁻¹ nm ⁻² ; r_0 0.58-0.70 nm
backbone angles	$\frac{1}{2} k (\theta - \theta_0)^2$	three consecutive residues	24	k 5-324 kJ mol ⁻¹ rad ⁻² ; θ_0 117-157°
backbone dihedrals	cosine ($n = 1$), damped when an angle approaches 180°	four consecutive residues	23	K 2.4-74 kJ/mol
stacking twist	cosine ($n = 1$)	B3-B5-B5-B3 of stacked guanines	8	K 391-774 kJ/mol (157-310 kT)
Morse within a tetrad	$D [1 - e^{-a(r-r_0)}]^2$	B3-B3, sides and diagonals (K_4)	18	r_0 0.38-0.65 nm; a 3.5-11.5 nm ⁻¹
Hoogsteen Morse	same	B2-B4 (N2-H...N7)	12	r_0 0.31-0.57 nm; a 6.1-11.0 nm ⁻¹
stacking Morse	same	B5-B5, same G-tract, adjacent tetrads	8	r_0 0.37-0.45 nm; a 5.9-8.9 nm ⁻¹
loop caps	same	loop/tail bases against the face of the outer tetrads	16	r_0 0.45-0.89 nm; a 2.3-5.1 nm ⁻¹
excluded volume	WCA, LJ 12-6 truncated and shifted at $2^{1/6}\sigma$	all pairs of types (1-2 and 1-3 excluded)	36 type pairs	ϵ 3.15 kJ/mol; σ 0.24-0.68 nm

All Morse terms have $D = 50$ kJ/mol (about 20 kT at 300 K), 2700 kJ/mol per copy in total: this is a native-state, Gō-like model, suited to 300 K but not to unfolding (see section 9).

2.3 PaiNN residual

U_{ML} is a sum of site energies, $U_{\text{ML}} = \sum_i \epsilon_i$, computed by a PaiNN network (polarizable atom interaction neural network, Schütt et al. 2021). The network is equivariant under rotations and invariant under translations and permutations of sites of the same type:

- **input:** site type (8 species $\rightarrow$ embedding of dimension $D = 64$) and positions. The graph includes all pairs of sites within $r_c = 1.2616$ nm, also between different copies (figure 1c);
- **radial basis:** 32 Gaussians centred between 0 and r_c , width $r_c/32$, multiplied by the Toxvaerd cutoff function $x^4/(x^4 + \alpha^4)$, with $x = (r_c - r)/r_c$ and $\alpha = 0.1$. The energy is smooth at r_c up to the third derivative;
- **interaction:** 2 message + update blocks on scalar features $s_i \in \mathbb{R}^{64}$ and vector features $v_i \in \mathbb{R}^{3 \times 64}$ (canonical PaiNN form, SiLU activations);
- **output:** MLP $64 \rightarrow 32 \rightarrow 1$ on the final scalar of each site. The value of the same species in isolation is subtracted from each ϵ_i , so that $U_{\text{ML}} \rightarrow 0$ for sites far from everything. The network has 106 049 parameters in total;
- **forces:** $F = -\nabla U_{\text{ML}}$ by automatic differentiation with respect to the site positions. On the guanines they are reduced to force and torque on the rigid body. The network runs in the ESPReso plugin in libtorch, FP32 on GPU, and the forces are conservative by construction.

Training: relative entropy, not force matching. The instantaneous AA forces projected onto the CG sites are dominated by noise: the learnable part is $R^2 \approx 0.5$ –1 %. Force matching on top of already fitted priors degraded the structure (intramolecular overlap from 0.976 to 0.94). The residual was therefore trained by minimising the relative entropy $S_{\text{rel}} = \langle \ln p_{\text{AA}}/p_{\text{CG}} \rangle_{\text{AA}}$, which uses positions only. The gradient is $\beta[(\partial U/\partial \theta)_{\text{AA}} - (\partial U/\partial \theta)_{\text{CG}}]$; the priors do not appear because they do not depend on θ .

- each iteration runs 25 ps of CG MD with the current model; the samples are reused by reweighting as long as the effective sample size per copy stays ≥ 50 %;
- the step is controlled by a trust region, and the last 20 % of the AA trajectory is held out as a check;
- **re0**: 30 iterations on top of the lp1 priors, starting from $U_{\text{ML}} \equiv 0$ (last output layer set to zero).
re1: 30 iterations on top of lp2 (lp1 plus the loop caps), starting from re0_it30. The production model is re1_it30;
- AA data: mapped positions of prod-1 (136 ns, 10 copies, one frame every 20 ps) at 300 K.

Effect of the residual on the intramolecular overlap (all sites), with the same protocol: lp2 priors only 0.980 $\rightarrow$ lp2 + re1_it30 0.995, against an AA vs AA ceiling of 0.995. The largest gain is in the B3-B3 channel (0.84 $\rightarrow$ 0.98); see section 3.

3. Intramolecular pair distributions $P(r)$

For each channel, $P(r)$ is the normalised histogram of the distances between sites of the same copy (same bins and normalisation as script 44). The channels separate the main contributions to the structure: B3-B3 for the Hoogsteen pairing in the tetrads, B5-B5 for tetrad stacking, S-S for the backbone; the "all sites" channel sums them, loops included. To isolate the contribution of the PaiNN residual we also compare a run with the **lp2 priors only**: same chain (2×20 ns), same initial state, same thermostat, PaiNN switched off.

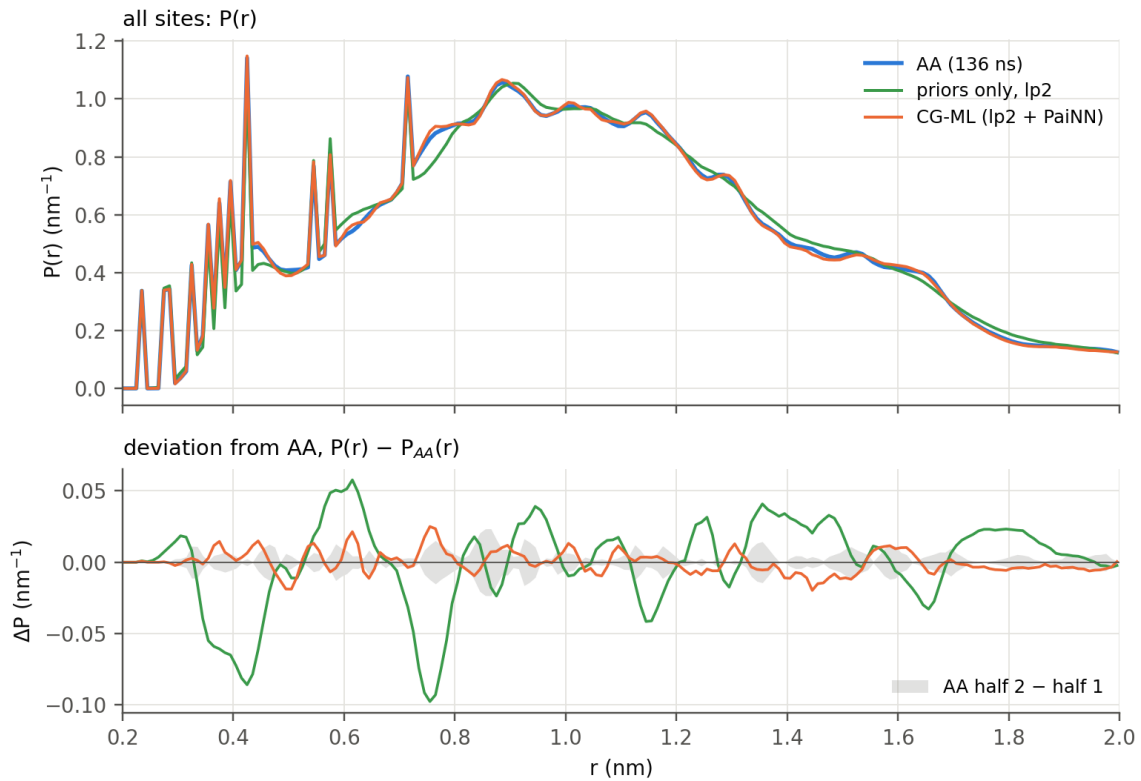


Figure 2. All-sites $P(r)$ (top) and deviation from the AA, $P(r) - P_{AA}(r)$ (bottom). Blue: AA; green: lp2 priors only; orange: lp2 + PaiNN. The grey band is the difference between the two halves of the AA trajectory, i.e. the noise of the reference. With the priors alone the deviation reaches $\pm 0.1 \text{ nm}^{-1}$ around 0.4, 0.6 and 0.75 nm (tetrad geometry and stacking) and stays systematic up to 2 nm; with PaiNN it falls within the band almost everywhere.

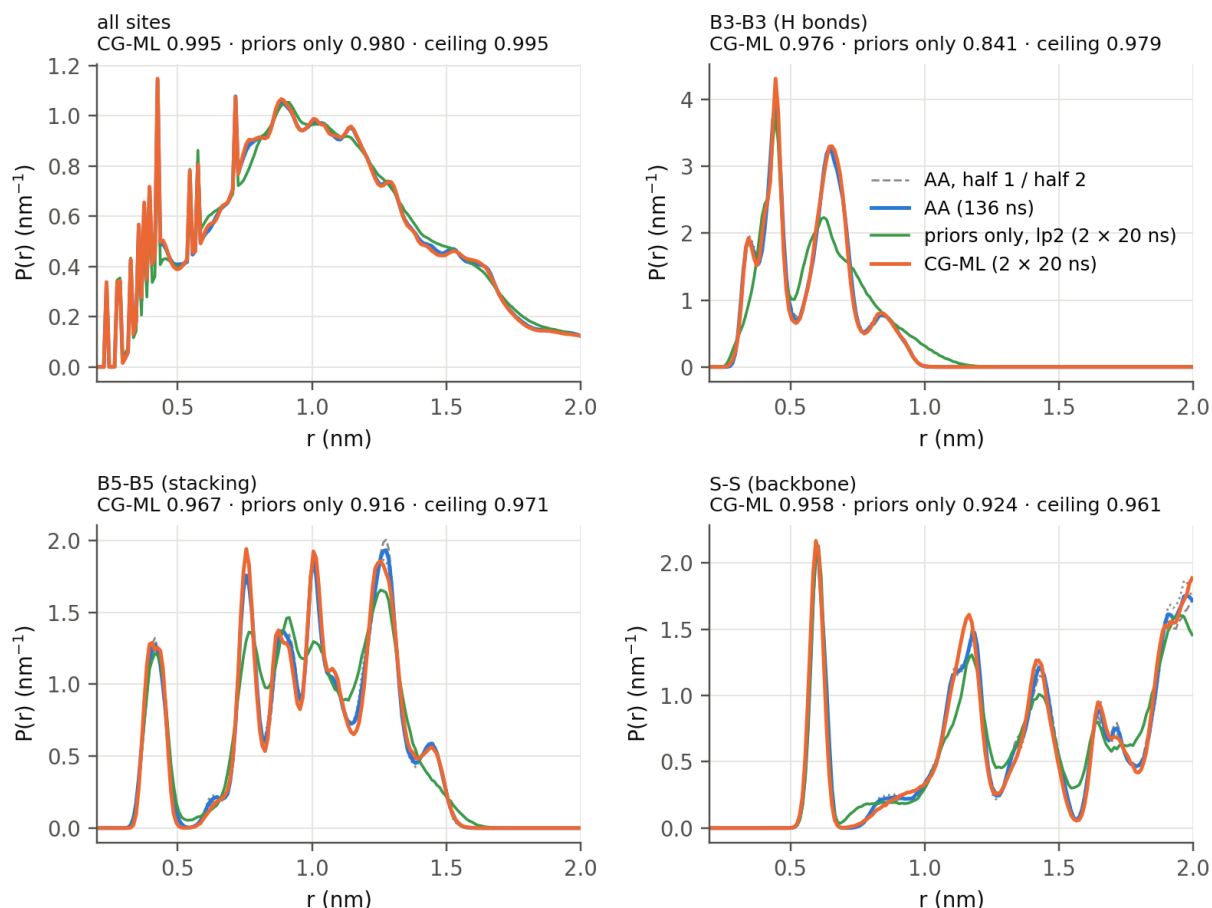


Figure 3. Intramolecular $P(r)$ per channel. Blue: AA over 136 ns; orange: CG-ML (lp2 + PaiNN) over 2 × 20 ns; green: lp2 priors only over 2 × 20 ns; grey dashed and dotted: first and second half of the AA. The titles report the overlaps with the AA and the ceiling (half 2 vs half 1).

channel	lp2 priors only	CG-ML (lp2 + PaiNN)	ceiling (AA half 2 - half 1)	CG-ML – ceiling	gap closed by PaiNN
all sites	0.980	0.995	0.995	-0.001	94 %
B3-B3 (H bonds)	0.841	0.976	0.979	-0.003	98 %
B5-B5 (stacking)	0.916	0.967	0.971	-0.004	92 %
S-S (backbone)	0.924	0.958	0.961	-0.003	91 %

PaiNN closes 91–98 % of the distance between the priors alone and the ceiling. The largest gain is in the B3–B3 channel: with the priors alone the second peak (tetrad diagonal, 0.65 nm) is low and broad (2.2 vs 3.3 nm⁻¹) and there is a tail up to 1.2 nm, i.e. distorted tetrads; with PaiNN both peaks return to the AA height. In the B5–B5 channel the priors blur the peaks at 0.75, 1.0 and 1.27 nm (relative orientation of the stacked tetrads), which PaiNN rebuilds.

In every channel the CG-ML reaches the ceiling: the gap is at most 0.004, within the variation between the two halves of the reference. Peak positions and heights coincide: for B3–B3 0.34, 0.445, 0.65 and 0.84 nm; for S–S the first neighbour at 0.595 nm. The remaining differences, for instance the S–S peaks at 1.17 and 1.42 nm slightly higher in the CG, have the same amplitude as the fluctuations between the AA halves. The geometry of the core (tetrads, stacking, backbone) is therefore reproduced at the level of distributions, not only on average.

4. $g(r)$ between different copies (short range)

The intermolecular $g(r)$ counts pairs of sites on different copies. Up to 2.0 nm, the largest distance computed, it describes only close encounters between copies: with $R_g \approx 0.8$ nm the copies rarely touch and $g(r)$ stays well below 1, in the excluded-volume region.

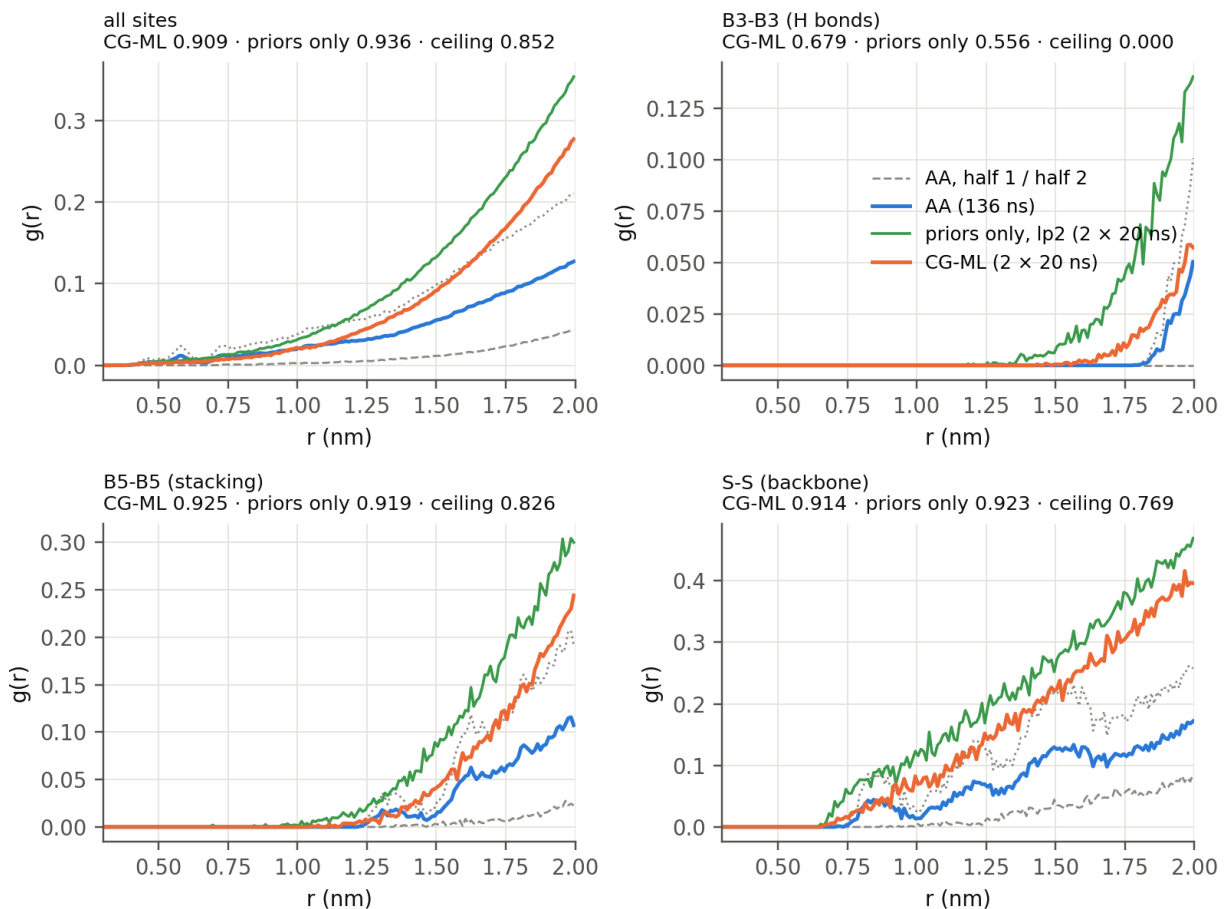


Figure 4. $g(r)$ between sites of different copies, same colours as figure 3. The two AA halves differ by a factor of ~ 5 : the encounters between copies in the AA are few long events.

channel	lp2 priors only	CG-ML vs AA	AA half 2 - half 1
all sites	0.936	0.909	0.852
B3-B3 (H bonds)	0.556	0.679	0.000
B5-B5 (stacking)	0.919	0.925	0.826
S-S (backbone)	0.923	0.914	0.769

In the AA the first half has almost no contacts between copies, the second half many more: these contacts are rare, non-ergodic events over 136 ns, and the reference does not sample them. The CG-AA overlap (0.68–0.93) exceeds that between the two AA halves (0–0.85), so here the metric does not discriminate. The CG-ML has somewhat more frequent contacts between copies than the AA average, similar to the second half. **Conclusion:** with this reference the short-range structure between copies cannot be verified. For the full intermolecular $g(r)$ (distances up to half the box) it is enough to rerun script 53 with a larger `--rmax`, but the AA uncertainty will remain the one set by 10 copies in a single box.

5. Radius of gyration

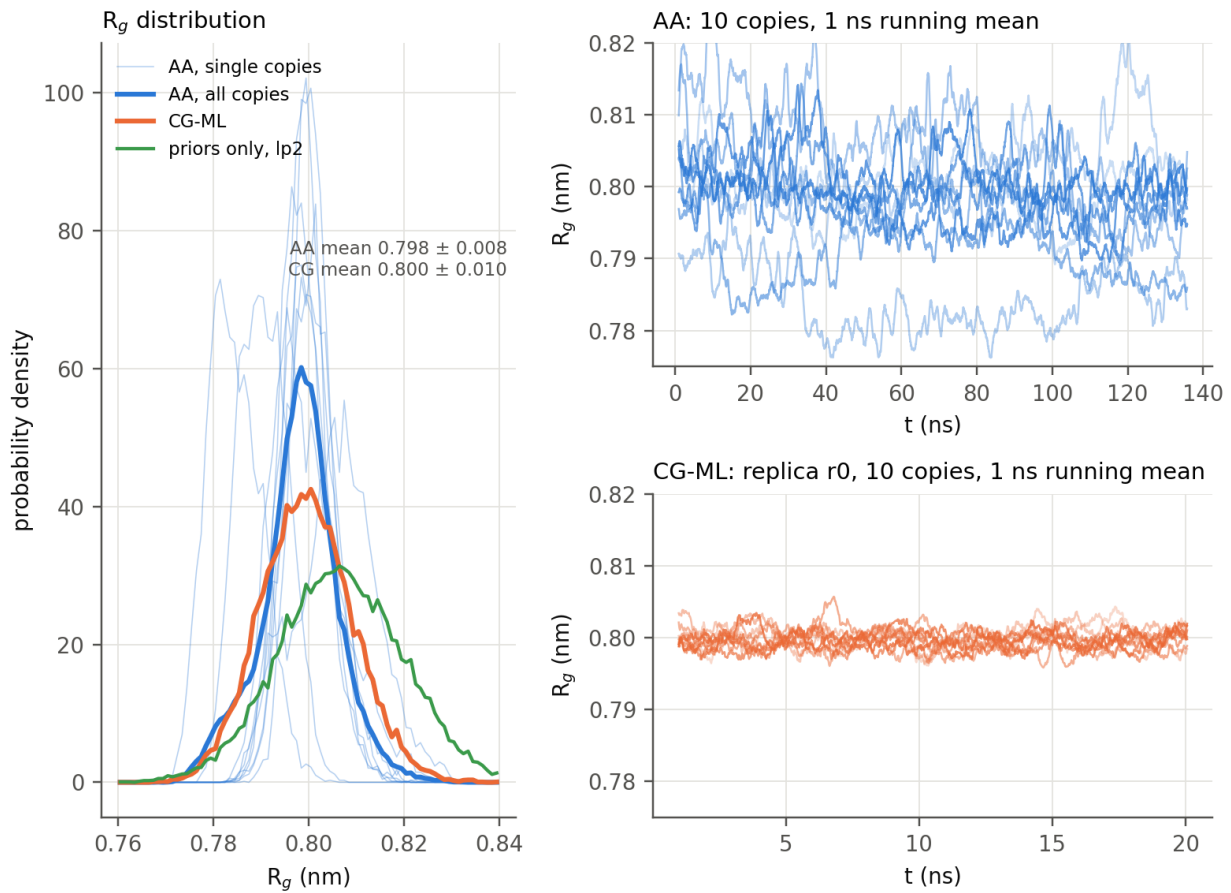


Figure 5. Left: R_g distribution (AA: single copies as thin lines, all copies as a thick line; CG-ML: all copies and replicas; green: lp2 priors only). Right: time series with a 1 ns running mean, AA over 136 ns and CG-ML (replica r0) over 20 ns, same vertical scale.

	AA	CG-ML	lp2 priors only
mean $\pm$ standard deviation (nm)	0.798 $\pm$ 0.008	0.800 $\pm$ 0.010	0.807 $\pm$ 0.013
means of the single copies (nm)	0.784-0.806	all $\approx$ 0.800	
τ at 1/e (script 47)	5.6 ns	2.4 ps	
overlap with the AA (ceiling)	-	0.86 (0.89)	0.62

The mean value is correct within 0.002 nm. The CG distribution is somewhat wider than the overall AA one and much wider than that of a single AA copy. The reason is in the time series: in the AA each copy stays for tens of ns on its own R_g (the most compact copy at 0.78 nm for most of the trajectory), with a slow 5–10 ns relaxation linked to loop rearrangements. In the CG the 1 ns average is flat and identical for all copies, because R_g decorrelates in a few ps. The CG thus samples a single "average" basin instead of the states that the AA visits slowly. For R_g the ratio of the τ values (~ 2400) measures different processes and is not a speed-up.

6. Other collective variables

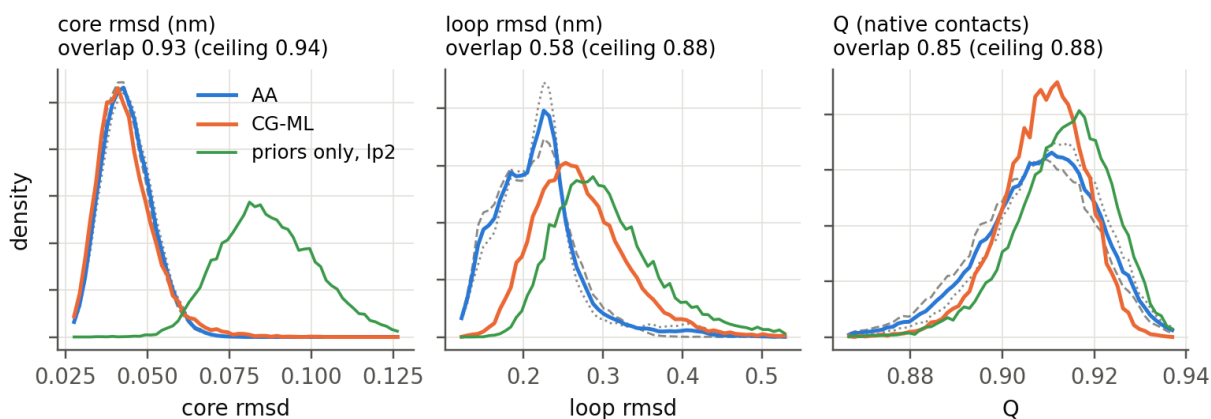


Figure 6. Distributions of core rmsd, loop rmsd and fraction of native contacts Q (grey: AA halves). Green: lp2 priors only. The titles report the CG-AA overlap and the ceiling.

variable	AA mean $\pm$ std	CG-ML mean $\pm$ std	overlap CG-ML	priors only mean $\pm$ std	overlap priors only	ceiling
R_g (nm)	0.798 ± 0.008	0.800 ± 0.010	0.86	0.807 ± 0.013	0.62	0.89
core rmsd (nm)	0.044 ± 0.008	0.044 ± 0.009	0.93	0.088 ± 0.015	0.04	0.94
loop rmsd (nm)	0.217 ± 0.053	0.272 ± 0.060	0.58	0.307 ± 0.069	0.39	0.88
Q	0.908 ± 0.013	0.909 ± 0.009	0.85	0.913 ± 0.011	0.82	0.88

With the priors alone the core rmsd is doubled (0.088 vs 0.044 nm) and its distribution barely overlaps the AA one (0.04): the tetrads stay formed (Q is even slightly higher) but are distorted and more mobile. PaiNN brings the core to the ceiling and also reduces the loop rmsd (0.31 $\rightarrow$ 0.27 nm).

The core is at the ceiling: rmsd 0.044 nm in both, overlap 0.93 vs 0.94. Q has the same mean but a somewhat narrower distribution in the CG (std 0.009 vs 0.013). The loops are the only clear discrepancy: in the CG they are more mobile and further from the mean structure (0.27 vs 0.22 nm), and the population at low rmsd (0.13–0.2 nm), which in the AA corresponds to the compact loop states, is missing. The diagnostics of scripts 49–52 attribute the gap to metastable states absent in the CG (state 1 of loop 3, minor state of loop 2) and to a partly wrong pairing register (T2–A15 in excess).

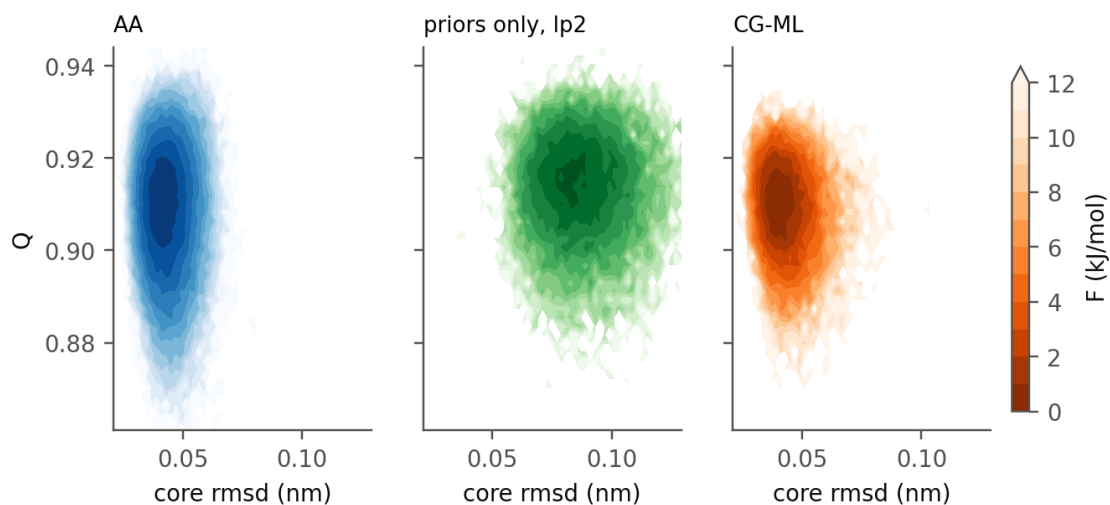


Figure 7. Free energy $F = -kT \ln p(\text{core rmsd}, Q)$: AA, lp2 priors only, CG-ML. The minimum of the priors only is shifted to rmsd $\approx$ 0.09 nm. The JSD between the AA and CG-ML FES (script 46) is 0.0787, equal to the AA vs AA ceiling (0.0786); the CG is noisier because of the smaller number of frames (20 080 vs 67 880 copy-frames).

7. Dynamics: translational and rotational MSD

MSD of the centre of mass of each copy and angular MSD of the core orientation (Kabsch alignment, rotations between successive frames summed as rotation vectors in the laboratory frame; script 48). The curves stop at a quarter of the length of each trajectory. In the bottom row the time axis is converted into GPU hours with the measured throughput.

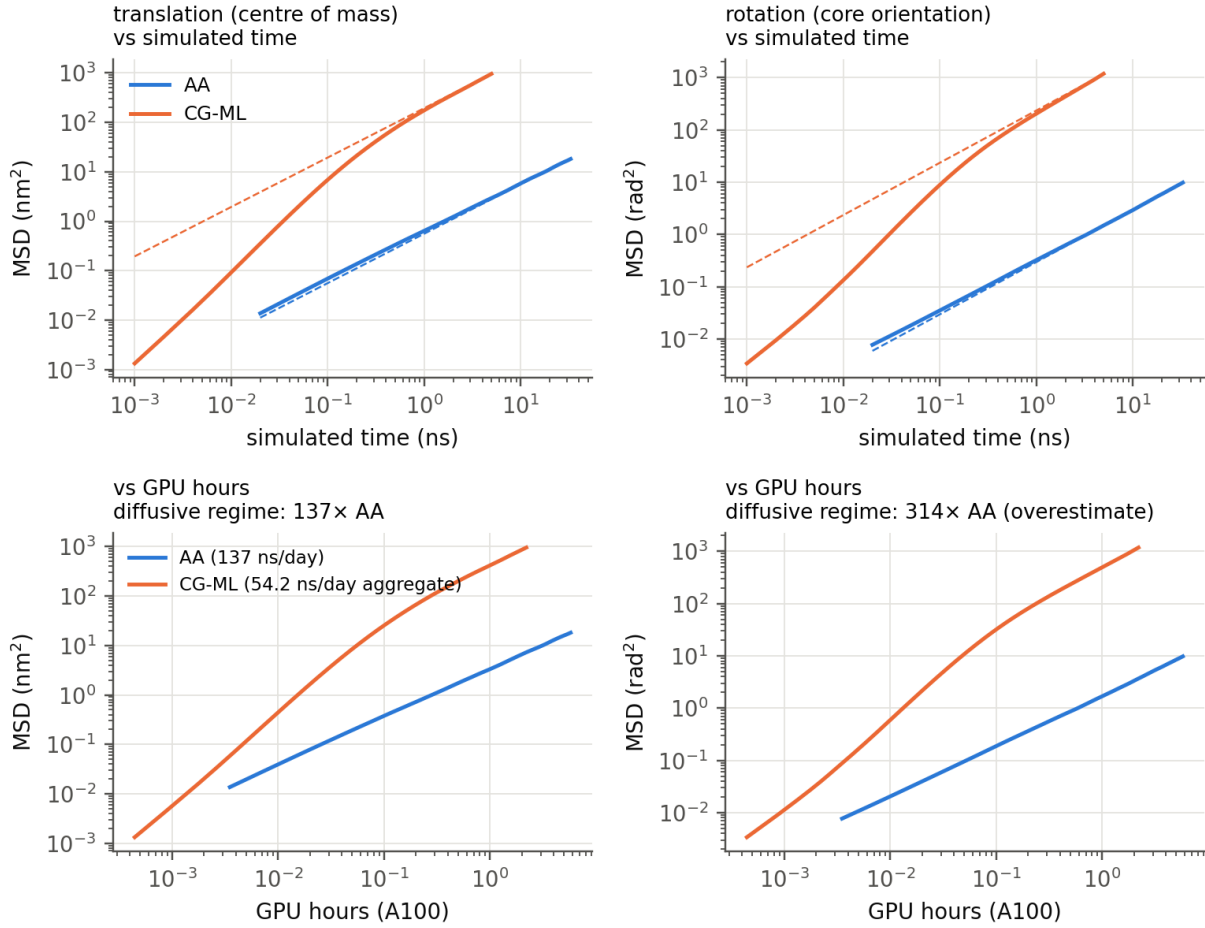


Figure 8. Top: MSD vs simulated time; dashed 6Dt with D from the linear regime. Bottom: the same curves vs A100 GPU hours (AA 137 ns/day; CG-ML 54.2 ns/day aggregate over 2 replicas).

	AA	CG-ML	CG/AA ratio
D (nm ² /ns), linear fit	0.093	32.4	347
τ_v from the Langevin fit (ps)	1.2	105	
translational MSD per GPU hour (nm ²)	3.19	439	137×
D _{rot} (rad ² /ns), linear fit	0.049	39.1	793
τ_{p2} (ps)	3313	29.8	$\alpha = 111$
1/(6D _{rot}) (ps)	3378	4.3	
angular MSD per GPU hour (rad ²)	1.69	530	314× (overestimate)
orientation decorrelation per GPU hour			44×

- **Translation:** in the CG the regime is ballistic up to $\sim 0.1\text{--}0.3$ ns ($\tau_v \approx 100$ ps), then diffusive; D is 0.68 of the value $kT/(N\gamma)$ expected for an isolated molecule. In the diffusive regime the MSD per GPU hour equals $6D \times (\text{ns per hour})$, i.e. $\alpha \times$ throughput ratio: 137×. With a few minutes of computation the CG is still ballistic and the gain is smaller.
- **Rotation:** in the AA $\tau_{p2} \approx 1/(6D_{\text{rot}})$, so rotation is diffusive and the method is confirmed. In the CG τ_{p2} is ~ 7 times $1/(6D_{\text{rot}})$: the orientation decorrelates already in the inertial regime (low γ),

and the angular MSD overestimates the gain in sampling orientations. For rotation the number to use is that of the decorrelation, 44×.

8. Efficiency per GPU

Efficiency = $\alpha \times (\text{ns/day}_{\text{CG}} / \text{ns/day}_{\text{AA}})$, with $\alpha = \tau_{\text{AA}}/\tau_{\text{CG}}$ (1/e times from script 47) or the ratio of the D values; $\text{ns/day}_{\text{CG}}/\text{ns/day}_{\text{AA}} = 54.2/137 = 0.396$.

observable	τ AA	τ CG-ML	α	efficiency per GPU
translation (MSD)			347 (ratio of D)	137×
core sites	175 ps	0.48 ps*	363*	~140×
rotation (τ_{p2})	3.31 ns	29.8 ps	111	44×
core rmsd	78.5 ps	1.10 ps	72	28× (minimum)
R_g , loops	5.6, 7.0 ns	2.4, 2.9 ps	~2400	not interpretable

* With one frame per ps the CG τ of the core sites (~0.5 ps) is not resolved (script 47 gives 0.69 ps by interpolation); τ and α are taken from the run sampled every 0.5 ps.

The number to quote is the minimum among the comparable observables: **~28× per GPU** (core rmsd), with translation and local core motions at ~140× and rotation at ~44×. Per node budget the advantage can grow up to ~4×, because the AA uses 32 cores against 8 for the CG; the AA throughput with 8 cores per GPU is needed to pin this down.

9. Limitations and caveats

- **Loops:** a single basin instead of the AA metastable states. State 1 of loop 3 (26 %) and the minor state of loop 2 are missing; the 5' tail is too mobile; the T2-A15 contact is in excess (0.56 vs 0.24 ± 0.09). State-dependent contacts (lp2c0) were tried and not adopted: with a single site for T and A the register specificity cannot be obtained with pair contacts.
- **AA reference not ergodic on the loops:** 17–61 % of the loop variance is between copies and the slow times are 4–10 ns over 136 ns. The AA populations of the loop states have large errors, and the ceilings on loops and R_g are optimistic: the two halves share the same stuck copies.
- **Native state only:** the 54 Morse terms at 50 kJ/mol per copy keep the quadruplex closed at any accessible temperature. With the priors alone, on one copy, no tetrad opens between 300 and 600 K in 250 ns, and the potential energy grows like that of a harmonic system. The model is meant for the structure and dynamics around the native state; for unfolding thermodynamics the contact depths must be recalibrated (work in progress, folder *tel26_unfold*).
- **g(r) between copies:** verified only at short range (≤ 2 nm), where the AA does not converge either.
- **Efficiency:** per A100 GPU, on the same hardware; the ratio per GPU hour depends on the node throughput, which varies by ~20 % between nodes with the same code.

Appendix: data provenance

Data: *report_data.npz* generated on Leonardo by *53_tel26_report_data.py* (P(r) and g(r) with the routines of script 44; collective variables with script 46; MSD curves from *msd_wall_prod.json*, script 48; times from *ts_prod.json*, script 47). AA reference: *tel26_lp1_dataset.bin*. CG-ML: *prod_re1_s0{1..4}_r{0,1}.samples.npz* (chain prod_re1, model lp2 + re1_it30); priors only: *pri_lp2_s0{1..4}_r{0,1}.samples.npz* (same chain with DISABLE_ML=1); data in *report_data_v2.npz*. Figures and PDF: *make_figs_en.py*, *build_report_en.py* and *model_fig/* in the folder *tutorials/tel26/report*. Parameters: P(r) with one AA frame every 2, CG every 20 ps, 200 bins up to 2 nm.